

BUSINESS INTELLIGENCE PROJECT REPORT

Forecast vs Actual

Financial Dashboard

Built with Microsoft Power BI

IT Budget Tracking · Variance Analysis · Department & Country Breakdowns · 2020 Full-Year View

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June 2025

Executive Summary

This report documents the design, development, and financial insights from an interactive Forecast vs Actual Financial Dashboard built in Microsoft Power BI. The dashboard analyzes 11,027 IT budget records spanning the full calendar year 2020 (January–December) across 38 IT departments, 25 cost element categories, and 18 countries.

This is the most financially sophisticated project in the portfolio — a Forecast vs Actual variance dashboard is the gold standard of financial planning & analysis (FP&A) reporting used by CFOs, budget managers, and IT directors worldwide. Building one in Power BI demonstrates mastery of financial data modeling, DAX variance calculations, and the ability to translate complex multi-dimensional budget data into a clear executive-level visual story.

Financial Snapshot — 2020 IT Budget

Metric	Value	Notes
Total Forecast (Full Year 2020)	\$261,051,603	Gross forecast across all depts, countries, cost elements

Metric	Value	Notes
Total Positive Forecast	\$277,532,873	Outflow / spending planned
Total Negative Forecast (Credits/Offsets)	-\$16,481,270	Inbound allocations, admin credits, offsets
Number of Forecast Records	11,027 rows	Monthly granularity across all dimensions
Peak Forecast Month	October 2020	\$24,348,063 — highest single month
Lowest Forecast Month	January 2020	\$17,585,796 — start-of-year ramp-up
Number of IT Departments	38	From Administration to Vendor Management
Number of Cost Elements	25	Labor, hardware, software, travel, legal, and more
Number of Countries	18	USA dominates at 83.4% of total budget
Largest Department by Budget	R2	\$49,362,667 — 18.9% of total forecast
Largest Cost Element	External Labor	\$80,819,314 — 31.0% of total
USA Share of Total Budget	83.4%	\$217.6M of \$261M total forecast
Tool Used	Microsoft Power BI Desktop	Interactive dashboard with slicers, KPI cards, variance charts

1. Introduction

1.1 Project Background

Every organization with a finance function produces a budget (forecast) at the start of each year and then tracks actual spending against that forecast throughout the year. The gap between forecast and actual — called variance — is the most important number in corporate financial management. Favorable variance means spending less than planned; unfavorable variance means overspending.

This Power BI dashboard was built to solve a real FP&A problem: how do you give a non-technical budget owner instant visibility into whether their department is on track, over budget, or under budget — broken down by cost category, country, and month? The answer is an interactive Forecast vs Actual dashboard that auto-updates as actuals are loaded.

1.2 What Makes This the Most Advanced BI Project

- Financial data modeling: Handling both positive (expense) and negative (credit/allocation) forecast values correctly

- Multi-dimensional analysis: 38 departments × 25 cost elements × 18 countries × 12 months = 205,200 possible combinations
- DAX variance calculations: Forecast – Actual, Variance %, cumulative YTD totals
- Dual-dataset architecture: Forecast table + Actual table joined on Date/Department/Country/Cost Element
- Hierarchical slicing: Filter from country → department → cost element → month for full drill-down
- FP&A-grade design: Matches the reporting standards used by Fortune 500 finance teams

1.3 Dashboard Objectives

- Display full-year 2020 IT forecast budget by department, country, and cost element
- Enable comparison of forecast vs actual spending at any level of the hierarchy
- Calculate and visualize variance (absolute and percentage) by dimension
- Provide monthly trend lines for both forecast and actual to show run-rate patterns
- Allow managers to slice and filter by any combination of IT department, country, cost element, and month
- Highlight departments or cost categories running significantly over or under budget

2. Dataset Overview

2.1 Data Source

The primary dataset is Forecast.csv — a structured IT budget forecast table covering January through December 2020. The dashboard design (shown in ForecastVsActualDashboard(p_2).pdf) incorporates both this forecast table and a companion Actual spending table, which together enable the variance analysis that is the dashboard's core purpose.

Repository: github.com/princemaheta2627-glitch/ForecastVSactual_dashboard_powerbi

2.2 Dataset Schema — Forecast.csv

Column	Type	Unique Values	Description
Date	Date	12	Monthly date in MM/YYYY format — Jan 2020 to Dec 2020
IT Dep.	Text	38	IT department name (e.g., R2, Manufacturing, Networking)
CostElement	Text	25	Spending category (External Labor, Software, Hardware, etc.)

Column	Type	Unique Values	Description
Country	Text	18	Country where the budget applies (USA dominates at 83.4%)
Forecast	Decimal	11,027	Budget forecast value in USD — includes negatives for credits/offsets

2.3 Full Dataset Statistics

Characteristic	Value	Interpretation
Total Rows	11,027	Each row = one forecast entry for a specific dept/cost/country/month
Date Range	Jan 2020 – Dec 2020	Full calendar year forecast — 12 monthly periods
Total Gross Forecast	\$261,051,603	Net budget after credits and offsets
Positive Forecast Sum	\$277,532,873	Planned spending across all categories
Negative Forecast Sum	-\$16,481,270	Credits, inbound allocations, admin offsets
Negative Value Rows	843	7.6% of rows represent credits or negative adjustments
Missing Values	None	All 5 columns fully populated — clean dataset
Forecast Range	-\$1,368,097 to +\$2,665,022	Wide value spread requiring balanced axis scaling

2.4 Monthly Forecast Distribution

Month	Forecast Total	MoM Change	Cumulative YTD
January 2020	\$17,585,796	—	\$17,585,796
February 2020	\$19,689,217	+11.9%	\$37,275,013
March 2020	\$21,660,830	+10.0%	\$58,935,843
April 2020	\$19,907,454	−8.1%	\$78,843,297
May 2020	\$21,931,683	+10.2%	\$100,774,980
June 2020	\$22,034,360	+0.5%	\$122,809,340
July 2020	\$23,311,753	+5.8%	\$146,121,093
August 2020	\$23,290,163	−0.1%	\$169,411,256
September 2020	\$23,561,677	+1.2%	\$192,972,933
October 2020	\$24,348,063	+3.3%	\$217,320,996

Month	Forecast Total	MoM Change	Cumulative YTD
November 2020	\$22,126,711	−9.1%	\$239,447,707
December 2020	\$21,603,896	−2.4%	\$261,051,603

Pattern: January is the budget ramp-up month (lowest at \$17.6M). July through October form the peak spending period, all exceeding \$23M/month. November–December show the typical year-end budget deceleration as projects close and funds are rebalanced.

3. Department & Cost Element Analysis

3.1 Top 10 Departments by 2020 Forecast Budget

Rank	Department	Total Forecast	% of Total	Category
1	R2	\$49,362,667	18.9%	Research/Engineering division
2	Manufacturing	\$40,066,799	15.3%	Production IT systems
3	Planning	\$22,972,297	8.8%	Strategic planning systems
4	R3	\$17,310,146	6.6%	Research/Engineering division
5	R5	\$16,671,636	6.4%	Research/Engineering division
6	Data Centers	\$15,149,959	5.8%	Infrastructure & hosting
7	Productivity	\$14,360,116	5.5%	Productivity tooling
8	Networking	\$14,187,876	5.4%	Network infrastructure
9	Hosting	\$9,518,574	3.6%	Managed hosting services
10	Core Infrastructure	\$8,868,519	3.4%	Core IT infrastructure

Top 3 insight: R2, Manufacturing, and Planning together represent 43% of the total IT forecast. The Research divisions (R1–R5) combined account for over \$86M — suggesting a major R&D-driven IT investment strategy. The smallest departments (Licensing, Other) are near-zero.

3.2 Cost Element Breakdown — Where the Budget Goes

Rank	Cost Element	Total Forecast	% of Total	Nature
1	External Labor	\$80,819,314	31.0%	Contractors, consultants, outsourced staff
2	Internal Labor	\$61,492,075	23.6%	FTE salaries allocated to IT
3	Amortization	\$50,887,542	19.5%	Asset cost spread (software/hardware)
4	Software Maintenance	\$23,511,610	9.0%	License renewal, SaaS subscriptions

Rank	Cost Element	Total Forecast	% of Total	Nature
5	Hardware	\$11,018,778	4.2%	Physical IT equipment purchases
6	Depreciation	\$10,768,579	4.1%	Asset value reduction accounting
7	Software	\$8,809,732	3.4%	New software purchases
8	Telecomm	\$5,617,476	2.2%	Telecommunications and connectivity
9	Other	\$5,501,543	2.1%	Miscellaneous unclassified spend
10	Administrative	−\$4,263,055	−1.6%	Net credit — administrative offsets

Critical finding: Labor (External + Internal combined) accounts for 54.6% of the total IT budget — \$142.3M of \$261M. This is the dominant cost driver and the most important line item for variance tracking. External labor alone exceeds internal labor by \$19.3M, indicating heavy reliance on contractors and consultants.

Amortization and Depreciation combined (\$61.7M) represent the non-cash accounting charges from prior capital investments — important for understanding CAPEX vs OPEX budget management.

3.3 Country Budget Distribution

Rank	Country	Total Forecast	% of Total	Region
1	USA	\$217,645,748	83.4%	North America — HQ dominant
2	United Kingdom	\$13,209,221	5.1%	Europe
3	France	\$6,304,010	2.4%	Europe
4	Puerto Rico	\$3,068,374	1.2%	Caribbean / US Territory
5	Germany	\$3,034,242	1.2%	Europe
6	Italy	\$2,772,417	1.1%	Europe
7	Belgium	\$2,264,183	0.9%	Europe
8	New Zealand	\$2,254,711	0.9%	APAC
9	Brazil	\$1,891,731	0.7%	South America
10	Spain	\$1,800,877	0.7%	Europe
11–18	Other 8 countries	\$6,806,089	2.6%	Mexico, Ireland, Canada, Switzerland, China, Austria, Japan, Netherlands

The USA accounts for 83.4% of the total IT budget — a strong US-centric operational footprint. The UK is the next largest at 5.1%, followed by a cluster of European countries in the 1–2.4%

range. APAC (New Zealand, Japan, China) and South America (Brazil, Mexico) have smaller but meaningful allocations.

3.4 Top Department–Country Combinations

Rank	Department	Country	Forecast	Strategic Meaning
1	R2	USA	\$46,254,320	Core US R&D engineering budget
2	Manufacturing	USA	\$30,231,939	US manufacturing IT systems
3	Planning	USA	\$22,456,690	US strategic planning tools
4	R5	USA	\$16,497,693	R5 engineering — US-based
5	R3	USA	\$16,290,060	R3 engineering — US-based
6	Data Centers	USA	\$12,094,878	US data center infrastructure
7	Networking	USA	\$10,648,166	US network infrastructure
8	Hosting	USA	\$9,409,659	US managed hosting
9	Productivity	USA	\$6,970,759	US productivity platform
10	Development	USA	\$5,028,365	US software development

The top 10 dept-country combinations are all USA-based, confirming the US-centric budget structure. The first non-US combination would appear around rank 20–30 in a full listing, likely UK-based departments.

4. Dashboard Design & Architecture

4.1 Power BI Data Model

The Forecast vs Actual dashboard uses a dual-table data model — the foundational architecture for any variance dashboard:

Table	Role	Key Columns	Join Type
Forecast	Budget plan table	Date, IT Dep., CostElement, Country, Forecast	Left table
Actual	Real spending table	Date, IT Dep., CostElement, Country, Actual	Right table
Date Table	Time intelligence	Date, Month, Quarter, Year	Shared dimension

The two tables are related via a shared Date dimension and matching dimension columns. This enables Power BI to calculate Forecast and Actual in the same visual context, making variance computation natural and dynamic.

4.2 DAX Measures — Variance Engine

Core Financial Measures

```
- Total Forecast =  
-     SUM(Forecast[Forecast])  
-  
- Total Actual =  
-     SUM(Actual[Actual])
```

Variance Measures (the core FP&A metrics)

```
- Variance (Absolute) =  
-     [Total Forecast] - [Total Actual]  
-  
- Variance % =  
-     DIVIDE([Variance (Absolute)], ABS([Total Forecast]), 0) * 100  
-  
- Favorable Variance Flag =  
-     IF([Variance (Absolute)] >= 0, "Under Budget", "Over Budget")
```

Time Intelligence Measures

```
- YTD Forecast =  
-     TOTALYTD([Total Forecast], 'Date'[Date])  
-  
- YTD Actual =  
-     TOTALYTD([Total Actual], 'Date'[Date])  
-  
- YTD Variance =  
-     [YTD Forecast] - [YTD Actual]
```

Month-over-Month Change

```
- MoM Forecast Change % =  
-     VAR PrevMonth = CALCULATE([Total Forecast],  
-                               DATEADD('Date'[Date], -1, MONTH))  
-     RETURN DIVIDE([Total Forecast] - PrevMonth, ABS(PrevMonth), 0) * 100
```


4.3 Power Query Transformations

Before building visuals, the Forecast.csv was processed in Power Query Editor:

- Date parsing: Converted DD/MM/YYYY text to proper Date type, set to first day of each month
- Forecast column: Removed embedded commas in quoted numbers ("-21,785" → -21785), cast to Decimal Number
- Column renaming: 'IT Dep.' → 'Department', 'CostElement' → 'Cost Element' for cleaner visual labels
- Trimmed whitespace: Department and Country name columns cleaned
- Negative value handling: Confirmed negative values are valid (credits, inbound allocations) — not errors
- Merged Actual table: Joined Actual dataset on [Date, Department, Cost Element, Country] using Left Outer Join

4.4 Dashboard Layout — Visual Architecture

Zone	Content	Visual Type
Top Header	'Forecast vs Actual — IT Budget 2020' title + period subtitle	Text Box
KPI Row (4 cards)	Total Forecast Total Actual Variance (\$) Variance (%)	Card × 4
Slicer Panel (left)	Department slicer Country slicer Cost Element slicer Month slicer	Slicer × 4
Main Chart (center)	Monthly Forecast vs Actual side-by-side + Variance line overlay	Combo Line + Clustered Bar
Waterfall Chart	Variance by Department — shows which depts are over/under	Waterfall Chart
Treemap	Budget allocation by Department (size = forecast amount)	Treemap
Donut Chart	Country share of total forecast	Donut Chart
Cost Element Bar	Horizontal bar — Top cost elements by total forecast	Bar Chart (horizontal)
Matrix Table	Dept × Month with Forecast, Actual, Variance columns	Matrix with conditional formatting
Variance Bars	Conditional formatting: green (under budget), red (over budget)	Data Bars + Color Rules

5. Key Financial Insights

5.1 Budget Seasonality & Cadence

The monthly forecast profile reveals a clear IT budget seasonality pattern consistent with enterprise financial planning cycles:

- January (\$17.6M): Lowest month — budget approvals finalized, projects starting up, hiring/contracting not yet at full pace
- February–March (\$19.7M–\$21.7M): Ramp-up as new-year projects launch and contractor onboarding completes
- April dip (\$19.9M): Minor pullback — possibly end-of-Q1 review and budget reforecast period
- May–October (\$21.9M–\$24.3M): Peak operational spending — projects in full execution, all resources engaged
- October peak (\$24.3M): Highest month — possible year-end project acceleration and Q4 hardware/software procurement before fiscal close
- November–December (\$22.1M–\$21.6M): Wind-down — project completion, budget freeze, reduced contractor utilization

Business Implication: The \$6.8M spread between January (lowest) and October (highest) represents a 38.7% swing in monthly spend. Any forecast-to-actual dashboard must account for this seasonality to correctly interpret monthly variances — a \$3M overspend in October may be normal; the same in January would be highly anomalous.

5.2 Labor Cost Concentration Risk

With 54.6% of the total budget in labor (External + Internal), the IT organization faces significant concentration risk:

- External Labor (\$80.8M) exceeds Internal Labor (\$61.5M) — more budget goes to contractors than to employees
- This creates flexibility (contractors can be reduced if forecasts miss) but also risk (contractor rates are more volatile)
- Any variance tracking system must prioritize the External Labor line — even a 5% overspend on this single line equals \$4M in unfavorable variance
- Labor lines are also the hardest to track precisely — hours × rates × location multipliers create complex actual calculations

Dashboard Value: The Forecast vs Actual dashboard makes labor variance immediately visible — managers can instantly see if contractor hours are running ahead of plan at the department level.

5.3 Geographic Budget Concentration

The 83.4% USA concentration means:

- Small exchange rate movements have minimal impact on the total forecast — a 10% USD/GBP shift only affects 5.1% of the budget
- US operations are the primary area for budget management attention

- International sites (especially UK, France, Germany) still warrant monitoring — \$23.4M combined in non-US budgets
- The dashboard's Country slicer allows a CFO to instantly isolate non-US spend and assess international over/under performance

5.4 R&D Investment Pattern

The R-series departments (R1, R2, R3, R5) combined represent \$85.1M — 32.6% of the total IT forecast. This suggests the organization is technology or pharma/biotech with heavy research IT infrastructure needs. The large R2 budget (\$49.4M) likely represents the core engineering platform supporting the research function.

5.5 Negative Budget Lines — Credit Management

843 rows (7.6%) have negative forecast values totaling −\$16.5M. These represent:

- Inbound Allocations: Costs recharged from another department — reduces the receiving department's net budget
- Administrative credits: Corporate overhead recovery entries
- The largest single negative value is −\$1,368,097 — a significant intra-company allocation

Dashboard consideration: The KPI cards must show NET forecast (positive + negative) to correctly reflect the true budget position. Displaying only positive values would overstate the budget by \$16.5M (6.3%).

6. Technical Skills Demonstrated

6.1 Financial Data Modeling

- Dual-table star schema: Forecast + Actual tables related through shared Date and dimension columns
- Understanding of CAPEX vs OPEX categories (Hardware/Software vs Labor/Maintenance)
- Handling negative budget values correctly in DAX measures (ABS() in variance %)
- Date table creation for time intelligence (TOTALYTD, DATEADD, SAMEPERIODLASTYEAR)

6.2 Advanced DAX

- TOTALYTD() — cumulative year-to-date aggregation respecting the current filter context
- DATEADD() — period-over-period comparisons (MoM, QoQ, YoY)
- DIVIDE() — safe division handling zero denominators in variance % calculation
- ABS() — absolute value for correct percentage calculation on negative budget lines

- IF() — conditional logic for favorable/unfavorable variance labeling
- VAR...RETURN — multi-step DAX calculations for MoM change measures
- CALCULATE() — the most powerful DAX function — overrides filter context for complex aggregations

6.3 Power Query Advanced Transformations

- Removing embedded commas from quoted numeric strings before type conversion
- Left Outer Join between Forecast and Actual tables on 4 matching keys
- Custom column creation: Month Name, Quarter, fiscal period labels
- Pivot/Unpivot operations if the source format required reshaping
- Error handling: replacing errors from invalid date or numeric parsing

6.4 FP&A Dashboard Design Principles

- Variance-first design: The most important number (\$ Variance and % Variance) appear in the top KPI row
- Conditional formatting: Red for over-budget, green for under-budget — instant visual signal without reading numbers
- Waterfall chart for variance: Shows which departments added or reduced total variance — the standard FP&A visualization
- Treemap for budget share: Immediately shows which departments dominate the total forecast
- Matrix with sub-totals: Enables drill-down from company total → department → cost element → country
- Combo chart (bar + line): Shows Forecast and Actual as bars, Variance as an overlay line — the classic FP&A trend view

7. Business Value & Use Cases

7.1 Who Uses This Dashboard

Stakeholder	Primary Question Answered	Dashboard Feature Used
CFO / Finance Director	Are we on track vs annual budget? What is full-year variance?	Top KPI cards + YTD variance
IT Budget Manager	Which departments are over/under budget this month?	Waterfall chart + Matrix table
Department Head (e.g., R2)	How is my department tracking vs forecast?	Dept slicer → personal KPIs

Stakeholder	Primary Question Answered	Dashboard Feature Used
Country Finance Manager	What is our country's IT spend vs plan?	Country slicer → variance cards
Procurement Team	Are hardware/software costs running over forecast?	Cost Element slicer → filtered view
External Auditor	Verify that actuals are supported and within approved budget	Full matrix with drill-down

7.2 Real Business Impact

- Early warning: A dashboard updated monthly catches overspend in month 3–4 rather than discovering it at year-end when it is too late to course-correct
- Accountability: Department heads can be held accountable to their specific budget lines with clear, undeniable data
- Reforecast trigger: If actuals deviate by more than 5% from forecast for 2 consecutive months, the dashboard automatically signals a reforecast requirement
- Audit readiness: The dashboard exports to PDF for budget review meetings and board presentations in seconds
- Labor arbitrage decisions: If External Labor is running 10% over plan, the manager can immediately see which departments are driving the overrun and take action (freeze new hires, renegotiate contracts)

8. Conclusion & Future Enhancements

8.1 Project Summary

The Forecast vs Actual Financial Dashboard is the most complex and business-critical project in the portfolio. It addresses a universal problem in corporate finance — budget variance management — and solves it with a professional, interactive Power BI solution that could be immediately deployed in any organization's finance function.

The dataset spans \$261M of IT budget across 38 departments, 25 cost elements, 18 countries, and 12 months — with both positive and negative values requiring careful handling. The resulting dashboard transforms this complexity into clear, actionable financial intelligence for non-technical stakeholders.

8.2 Future Enhancements

Enhancement	Description	Technique
Rolling Forecast	Replace static annual forecast with monthly reforecast updates	CALCULATE + USERELATIONSHIP
Variance Drill-Through	Click a department KPI card to open a full detail page	Power BI Drill-Through pages

Enhancement	Description	Technique
Budget Alert System	Email alert if variance exceeds $\pm 10\%$ for any department	Power BI Alerts + Power Automate
3-Year Trend View	Compare 2020 vs 2021 vs 2022 forecast performance	SAMEPERIODLASTYEAR DAX + multi-year data
CAPEX vs OPEX Split	Separate capital vs operating budget tracking	Cost Element classification column
Headcount Integration	Link labor budget to headcount data for per-FTE cost analysis	People data join
Currency Conversion	Apply live FX rates to convert country budgets to USD base	DAX currency parameter table
Predictive Forecast	Use ML to predict year-end actual based on current run-rate	Azure ML integration in Power BI

8.3 Learning Outcomes

This project was transformative in developing financial domain knowledge alongside Power BI technical skills. Understanding the difference between gross and net forecast, how inbound/outbound allocations work, why amortization and depreciation appear in IT budgets, and how to design a variance waterfall chart are all skills that bridge data science with financial analysis. The dual-table data model — joining Forecast and Actual tables — is the architectural pattern used in virtually every enterprise BI system, and mastering it in Power BI provides a foundation for more advanced data warehouse and SSAS work.

9. Project Information

Project Title	Forecast vs Actual Financial Dashboard — Power BI
Author	Prince Maheta
GitHub Repository	github.com/princemaheta2627-glitch/ForecastVsActual_dashboard_powerbi
Dashboard File	ForecastVsActualDashboard(p_2).pdf (exported from Power BI Desktop)
Data File	Forecast.csv — 11,027 rows $\times$ 5 columns (+ Actual table in Power BI model)

Financial Scope	\$261,051,603 total IT forecast — Jan 2020 to Dec 2020
Dimensions	38 IT Departments × 25 Cost Elements × 18 Countries × 12 Months
Primary Tool	Microsoft Power BI Desktop
Secondary Tools	Power Query Editor (data prep + table join), DAX (variance & time intelligence)
Key DAX Functions	TOTALYTD, DATEADD, DIVIDE, ABS, IF, CALCULATE, VAR/RETURN, SAMEPERIODLASTYEAR
Project Type	Financial Planning & Analysis (FP&A) BI Dashboard
Date Completed	June 2025
Report Date	June 7, 2026

This report was prepared as part of a data science & business intelligence portfolio.
Prince Maheta | Data Science Portfolio | 2025